

Eight Structural Principles for Adaptive AI Architecture

From Physical Law and Cybernetics to Engineering Constraints on Viable Adaptive Systems

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~MILO™ — Modular Intelligent Learning Orchestrator (patent pending)

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Abstract

Adaptive artificial-intelligence systems are described in vague terms — *self-improving, resilient, agentic* — that obscure the structural constraints under which such systems can actually be viable. This paper identifies eight principles that bound the design space of adaptive AI architectures. Six are established physical and informational laws — the Second Law of Thermodynamics, Ashby's Law of Requisite Variety, Shannon's information theory, the Principle of Least Action, Lyapunov stability, and the power-law distribution of system events. Two are original frameworks proposed by the author for the operator-cognitive performance layer of high-consequence systems: *Individual-Baseline Variance Modeling* and *Precision Perturbation Without Variance Compression*. The eight principles operate as architectural design constraints, not as a theory of intelligence. The first six are illustrated with their operational form inside MILO, a patent-pending adaptive AI orchestrator developed by the author and submitted to the U.S. Department of Energy under the Genesis Mission; the seventh and eighth are v.5 design-stage frameworks, not shipped operator-monitoring features. The synthesis is consistent with, and complementary to, Friston's Free Energy Principle, Beer's Viable System Model, and recent thermodynamics-adjacent AI work; it differs in that the eight principles are used as design-time constraints on architectural choices, not as a unified explanation of intelligence.

Keywords: adaptive AI, architectural constraints, cybernetics, Lyapunov-style bounded response, antifragility, human-in-the-loop, industrial AI orchestration.

Highlights.

- Identifies eight structural principles that bound the design space of viable adaptive AI architectures, drawn from established physical, informational, control-theoretic, and statistical lineages.
- Six principles apply established external laws (Second Law of Thermodynamics, Ashby's Law of Requisite Variety, Shannon Information Theory, Principle of Least Action, Lyapunov-style bounded response, Power-Law distribution).
- Two original frameworks proposed by the author for the operator-cognitive performance layer: *Individual-Baseline Variance Modeling* and *Precision Perturbation Without Variance Compression* — flagged as design-stage, pending empirical validation.

- Synthesis is positioned as complementary to Friston's Free Energy Principle and Beer's Viable System Model but distinct — design-time constraints on architectural choices, not a unified theory of intelligence.

Index Terms: adaptive AI architecture, structural principles, cybernetics, control theory, Lyapunov stability, Ashby's Law of Requisite Variety, Shannon Information Theory, antifragility, Power-Law distribution, operator-cognitive modeling, viable system model.

Plain Language Summary. Adaptive AI systems are often described in vague terms — *self-improving, resilient, agentic* — that obscure what makes some such systems viable and others fragile. This paper identifies eight engineering principles that constrain the design space of viable adaptive AI architectures. Six are established physical and informational laws (thermodynamic entropy, Ashby's variety law, Shannon information theory, the principle of least action, Lyapunov-style bounded response, and the power-law distribution of system events) applied here as architectural design constraints rather than as theories of intelligence. Two are original frameworks proposed by the author for the operator-cognitive performance layer of high-consequence systems: *Individual-Baseline Variance Modeling* and *Precision Perturbation Without Variance Compression*. The eight principles, taken together, separate adaptive AI orchestrators that survive operational stress from those that fail it.

Relevance to U.S. National Interest. The principles articulated here apply directly to the AI-enabled critical-infrastructure environments identified by the DOE Genesis Mission — advanced manufacturing, grid reliability, autonomous systems, nuclear-facility operations, and human-in-the-loop AI for high-consequence decision support. Adaptive AI deployed in those environments without these architectural constraints carries failure modes that the constraints were established to prevent.

Status of claims. Six of the eight principles in this paper apply established physical and informational laws as architectural design constraints: the Second Law of Thermodynamics, Ashby's Law of Requisite Variety[6], Shannon Information Theory[7], the Principle of Least Action, Lyapunov stability[8], and the Power-Law distribution of complex-system events [9]. The mapping of each principle to a specific orchestrator-level architectural choice is the author's contribution and is subject to further validation; the laws themselves are external references. Principles 7 and 8 — *Individual-Baseline Variance Modeling* and *Precision Perturbation Without Variance Compression* — are original frameworks proposed by the author for the operator-cognitive performance layer of high-consequence systems and are flagged in the paper as design-stage frameworks pending empirical validation in shipped deployments. The synthesis is consistent with prior work by Friston[3], Beer [4], and the thermodynamic-AI literature[5]; differences are explicit in §2 and §5. This manuscript is a preprint prior to peer review.

About MILO. MILO (Modular Intelligent Learning Orchestrator) is a patent-pending adaptive AI orchestration architecture for high-consequence critical-infrastructure environments. The full architectural reference — eight structural principles, eight operational integrity constraints, the unifying viability principle, and the trademark/patent status — is maintained as a single canonical document at <https://github.com/jmontano1/milo-architecture> (concept DOI: 10.5281/zenodo.20117025). Author contact: jmontano@jmautomated.com.

1. Introduction

Adaptive AI architecture is most often described in language drawn from the system's marketing rather than its physics. A system is *self-improving* because its developers say so; it is *resilient* because it has not yet failed visibly; it is *agentic* because it issues commands. These descriptors are claims about capability, not about structural form. They do not tell an engineer whether the system can survive a perturbation it was not trained for, whether its internal variety matches the variety of the environment it governs, whether its informational paths admit auditable decisions, or whether its adaptation has any bounded stop condition.

The history of engineering offers a better discipline. Physical and informational systems are not viable because they are aspirationally well-intentioned; they are viable because their architecture obeys constraints that have been established for decades — in thermodynamics, in classical mechanics, in cybernetics, in control theory, in information theory, and in the empirical statistics of complex systems. An adaptive AI architecture that ignores those constraints inherits failure modes the constraints were discovered to prevent.

This paper identifies eight such principles and argues that they constitute structural constraints on the design space of viable adaptive AI systems. Six are established laws applied as architectural design constraints. Two are original frameworks proposed by the author for the operator cognitive performance layer of high-consequence industrial systems, where human decision context, cognitive load, fatigue, and task-state misalignment can become material sources of operational variance.

The eight principles are illustrated using MILO (Modular Intelligent Learning Orchestrator), a patent-pending adaptive AI orchestration system[1] that the author has developed and submitted to the U.S. Department of Energy under the Genesis Mission[2]. MILO is the working substrate for the first six principles and the design substrate for the proposed v.5 operator-layer extension; the paper's contribution is the principles, not the implementation. Where MILO is referenced, it is referenced at the architectural level only.

The paper does not propose a new theory of intelligence. Several frameworks — Friston's Free Energy Principle and Active Inference[3], Beer's Viable System Model [4], and thermodynamic-adjacent AI explanation work [5] — already address broader explanatory or diagnostic layers. The contribution here is narrower and more practical: a set of design-time architectural constraints that, taken together, separate viable adaptive AI orchestrations from systems that merely claim viability.

2. Background and Related Work

The eight principles draw from five established intellectual lineages. The Second Law of Thermodynamics, formalized by Clausius and refined by Boltzmann, established entropy as the directional quantity governing closed-system evolution. Ashby's Law of Requisite Variety[6], introduced in 1956 cybernetics, established that a regulator must possess variety at least equal to the system it regulates. Shannon's information theory[7], introduced in 1948, established the channel-capacity bounds on noise-tolerant communication. The Principle of Least Action, developed by Maupertuis, Lagrange, and Hamilton, established that physical systems traverse trajectories of minimum action. Lyapunov stability[8], introduced in 1892, provided the formal mathematical condition for bounded return-to-equilibrium under perturbation. Power-law statistics in complex systems, formalized in modern form by Clauset, Shalizi, and Newman[9], established that heavytailed empirical distributions dominate the consequence-profile of complex-system events.

These five lineages have been applied to AI in adjacent but distinct ways. Friston's Free Energy Principle[3] proposes a unified explanation of self-organizing systems combining information-theoretic and thermodynamic primitives. Beer's Viable System Model [4] applies Ashby's variety law and cybernetic feedback to recursive viable organizations and autonomous systems.

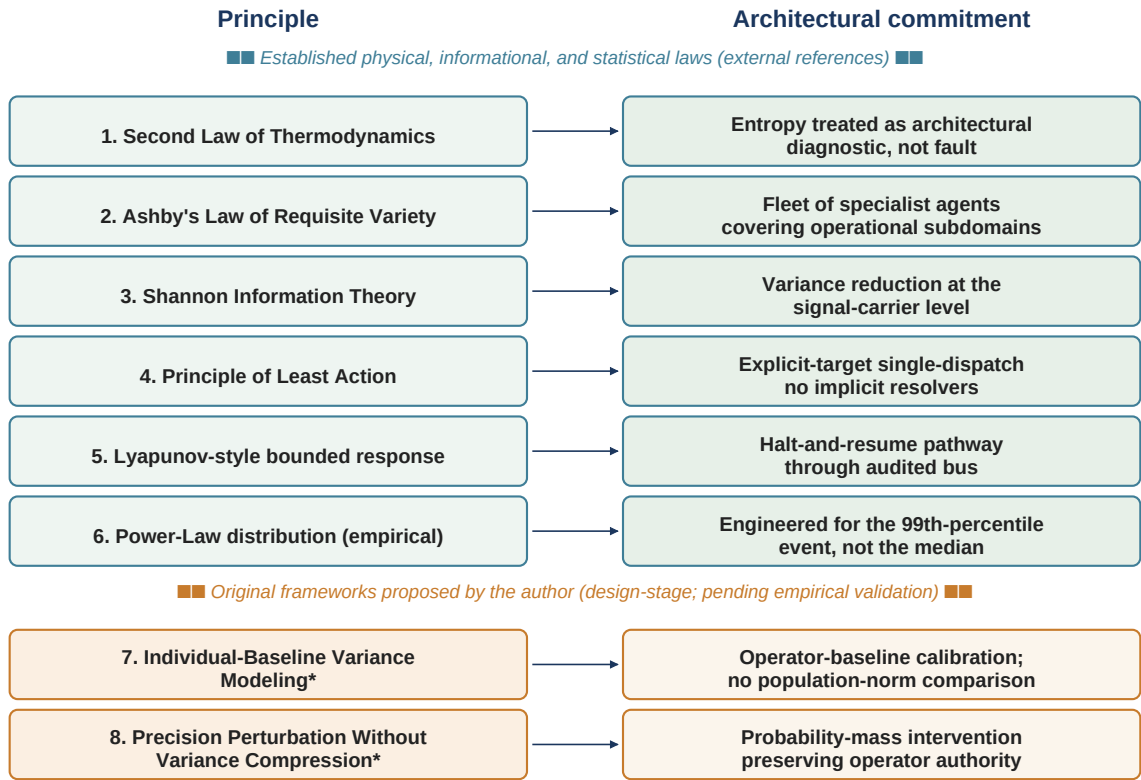
Thermodynamics-inspired AI work [5] has recently used thermodynamic concepts to explain black-box model behavior. Antifragility, introduced by Taleb [10], formalizes the inverted Jensen inequality under heavy-tailed payoff distributions and provides a structural account of systems that benefit from bounded disorder, variation, and stressors.

The contribution of the present paper differs from each of these. It is not a theory of intelligence (as in [3]), not a replacement for recursive viable-system cybernetics (as in [4]), not a thermodynamic explanation of AI behavior (as in [5]), and not a payoff-distribution analysis (as in [10]). It identifies the architectural choices an adaptive AI orchestration system must make and shows that each of the eight principles constrains those choices. The contribution is at the level of engineering design, not at the level of theoretical foundation. The two original frameworks (Sections 3.7 and 3.8) extend the synthesis into the operator-cognitive performance layer — a domain where prior work in human factors and adaptive automation[11], [12] has identified the problem but has not articulated this architectural form of the solution.

3. The Eight Structural Principles

The eight principles are organized in two groups. Sections 3.1 through 3.6 present the six established laws applied as architectural constraints. Sections 3.7 and 3.8 present two original frameworks proposed by the author for the operator-cognitive performance layer. Each principle is stated, its architectural form is identified, and the failure mode it prevents is named.

■ Eight structural principles → architectural commitments



* Original frameworks introduced for the operator-cognitive performance layer; full operational specification is forthcoming work.

principles to their architectural commitments. The top six are established external laws applied as design constraints; the last two are original frameworks proposed by

Selection criterion. The eight principles are selected against two filters. (i) The principle has a well-developed mathematical or empirical foundation in its native field — thermodynamics, cybernetics, information theory, classical mechanics, control theory, complex-systems statistics, or (for Principles 7 and 8) cognitive science. (ii) The principle has an operational architectural form: a design choice in an adaptive AI orchestration system whose specification can be falsified by observing the system's runtime behavior. Other foundational principles — conservation laws, symmetry principles, dissipation theorems — satisfy filter (i) but, in the author's assessment, are difficult to operationalize as architectural design constraints at the orchestration scale. The eight selected are not claimed to be the complete set of foundational principles applicable to adaptive AI; they are the set for which the author can articulate both the foundation and the architectural form. The framework is open to extension if additional principles meeting both filters are subsequently identified.

Scope and intent of this paper. The present paper is a *perspective paper* — a synthesis and map across foundations, not a deep treatment of any single principle. Each of the eight principles merits its own dedicated technical paper that develops the principle's operational architectural form in greater mathematical and empirical detail than space permits here. The contribution of the present paper is the synthesis: the identification of which principles bound the design space of adaptive AI orchestration and how each maps to an operational design choice. Readers seeking deeper treatment of any individual principle should expect that to be the subject of subsequent work.

Orthogonality and overlap among the principles. The eight principles are not strictly orthogonal in the mathematical sense; some derive from or are formally related to others. Most notably, Ashby's Law of Requisite Variety (Principle 2) can be derived from Shannon's Theorem 10 [7], making Principles 2 and 3 formally related rather than independent. Lyapunov-style bounded response (Principle 5) and the thermodynamic-entropy diagnostic (Principle 1) share the structural concern of bounded behavior under perturbation, viewed from different physical primitives. The principles are presented as eight because each has a distinct operational architectural form even where the underlying foundations are interrelated; the architecture of an adaptive AI orchestrator that satisfies all eight is structurally over-determined in a deliberate way, with overlapping constraints reinforcing one another rather than redundantly stating the same requirement.

3.1 Principle 1 — Second Law of Thermodynamics: Entropy as Architectural Diagnostic

The Second Law establishes that closed systems trend toward maximum entropy. Applied as an architectural constraint, the principle requires that entropy be treated as a *diagnostic signal* about system state, not as a fault condition to be suppressed. Architecturally, this implies modular construction in which each component can be observed for entropy drift independently, and in which any component can be replaced without systemic collapse. In MILO, security drift is detected by an integrity-monitoring subsystem, carried by the audit-first signal substrate, and addressed by a bounded incident-response pathway — entropy is observed and responded to as information about state. The architectural consequence is that no subsystem may be a single point of failure; the failure mode prevented is the monolithic redeploy required to repair a single drifted component.

3.2 Principle 2 — Ashby's Law of Requisite Variety

Ashby's law states that a regulator must possess variety at least equal to the variety of the system it regulates[6]. Applied as an architectural constraint, the principle requires that an AI orchestrator enumerate response options spanning the operational variety it is meant to govern. Implementation by a single generalist agent is structurally insufficient when the operational domain admits distinguishable subdomains; the orchestrator must instead operationalize a fleet of specialist agents whose combined variety matches the domain. In MILO, a fleet of specialist agents — each registered with a single explicit role covering an operational subdomain — provides the requisite variety against the domains the orchestrator governs; adding variety is structurally simple, since adding an agent is one new fleet entry while routing changes remain localized. The failure mode prevented is the AI orchestrator with five canned responses attempting to govern a domain of fifty distinguishable states.

3.3 Principle 3 — Shannon Information Theory: Variance Reduction at the Architectural Level

Shannon's information theory bounds the noise-tolerant capacity of any communication channel [7]. Applied as an architectural constraint, the principle requires that variance reduction occur at the carrier level — the signal infrastructure itself — rather than being redundantly implemented at each consumer. Architecturally, this implies a persistent, fanout-aware signal bus that admits reflex-level interception before subscriber delivery. In MILO, the signal substrate persists each signal, evaluates reflex predicates, then fans out to subscribers; reflex arcs run before fanout, short-circuiting high-severity events at the carrier level. The failure mode prevented is every consumer reinventing its own noise filter and disagreeing about what is signal.

3.4 Principle 4 — Principle of Least Action: Single-Target Dispatch

The Principle of Least Action establishes that physical trajectories minimize the action integral. Applied as an architectural constraint, the principle requires that the orchestration path between origin (the human operator's intent) and target (the responsible component) admit minimum informational cost — concretely, no hidden routing layer, no implicit resolver, no opaque dispatcher. In MILO, every command has one explicit target; the command bus persists the command, looks up the target, and invokes the single registered handler. The architectural patterns are *explicit-target dispatch* and *persist-before-deliver*: one command, one target, audited before dispatch [1]. The failure mode prevented is command routing through an opaque resolver where failures cannot be traced to a specific dispatch — the failure mode the author observed in pre-MILO orchestration that motivated the rebuild.

3.5 Principle 5 — Lyapunov-Style Bounded Response

Lyapunov's foundational work on stability[8] provides the formal mathematical condition for bounded return-to-equilibrium under perturbation. The architectural application here is the weaker but operationally important *Lyapunov-style bounded response*: the principle requires that any adaptive subsystem admit a bounded response to disturbance — concretely, an explicit halt-and-resume pathway that can be invoked when the subsystem departs its equilibrium zone — without claiming a formal Lyapunov-function analysis of the orchestrator's full state space. In MILO, an emergency-halt reflex detects critical signals, dispatches a halt command through the command bus, and symmetrically supports a resume command; every halt and resume traverses the same audited dispatch path as any other command, with end-to-end audit entries from critical signal through reflex through halt-executor. The architectural consequence is that adaptation which drifts unboundedly is not learning, it is failure; the failure mode prevented is positive-feedback runaway in an "adaptive" loop that has no architectural stop condition.

3.6 Principle 6 — Power-Law Distribution Architecture: Tail-Event Preparedness

Principle 6 differs from Principles 1 through 5 in epistemic status. Heavy-tailed distributions are an *empirical regularity* observed across many complex-system domains rather than a derived physical or mathematical law in the strict sense. Inclusion here is justified under filter (ii) of the selection criterion above: heavy-tailed consequence profiles in adaptive AI orchestration are an architectural design constraint with operational form (engineered for the 99th-percentile event), regardless of whether the underlying empirical regularity rises to the status of a law in the philosophical sense.

Empirical statistics in complex systems show that heavy-tailed distributions dominate consequence profiles[9]. Applied as an architectural constraint, the principle requires that an adaptive AI orchestrator be engineered for the 99th-percentile event, not the median. Architecturally, this implies rolling-window degradation detection (rather than threshold-only alarms), periodic self-monitoring on a bounded cadence (rather than operator-triggered checks), and bounded reporting (top-N source ranking) to prevent flooding under tail events. In MILO, an integrity-monitoring subsystem implements all three: a health monitor with rolling-window error-and-critical detection per source, a periodic self-monitoring scheduler on a bounded cadence, and a signal aggregator that ranks sources by cumulative count without flooding the dashboard. The architectural consequence is design for the tail, not the median; the failure mode prevented is the system that meets 50th-percentile SLO and catastrophically fails at the 99th.

This principle invokes the related concept of *antifragility* [10]: bounded variation and disorder that can improve a system rather than merely degrading it. The present framing treats antifragility as an architectural property of the orchestrator's adaptive-learning loop, not as a generalized claim about humans inside the system; the application of antifragility-style framings to operators is addressed in Section 3.8 and Section 4 with explicit governance constraints.

3.7 Principle 7 — Individual-Baseline Variance Modeling (Original Framework)

Status — original framework proposed by the author. Sections 3.7 and 3.8 introduce frameworks proposed by the author for the operator-cognitive performance layer of high-consequence adaptive AI orchestration systems. These frameworks have been submitted to the U.S. Department of Energy under the Genesis Mission[2] and are under active development. The discussion below is at the principle level.

Established human-factors research has long documented that human error and operator performance vary across shifts, roles, environmental conditions, training histories, and task contexts [13], [14]. The original framework proposed here is the architectural application: an adaptive AI orchestration system that intervenes on the operator-cognitive performance layer must model variance against the individual operator's own established performance baseline, not against a population norm. The architectural form of the principle is: *Interventions are calibrated to the individual's measured deviation from their own optimal performance state, established over a defined baseline window and recalibrated on a defined cadence.* The failure mode prevented is the AI system that misfires on operators whose individual baseline differs legitimately from a population norm.

The framework operates under the operational integrity constraints of *individual consent*, *individual-baseline-only measurement*, *no surveillance architecture*, and *operator authority as the architectural invariant* [1]. Implementation in operational systems requires institutional ethics review, published consent frameworks, and periodic third-party audits of system influence patterns and operational outcomes.

3.8 Principle 8 — Precision Perturbation Without Variance Compression (Original Framework)

Status — original framework proposed by the author. Established cognitive-science and human-reliability work supports probabilistic modeling of human state, workload, and decision reliability [3], [12]. The original framework proposed here is the architectural intervention category: *precision perturbation* — a class of intervention that shifts probability mass in the operator's cognitive state distribution toward high-reliability decision outputs without overriding operator authority and without compressing the essential variability required for adaptive operational judgment.

The framework is the explicit architectural inverse of two failure modes commonly observed in operator-facing AI systems: (a) override-style interventions that bypass operator authority and (b) compression-style interventions that drive operators toward homogeneous decision states, eliminating the variability that is the operator's adaptive intelligence. The architectural form of the principle is: *Interventions are calibrated as precision perturbations to the operator's probabilistic cognitive state — neither override nor compression — preserving both authority and variability.*

This framework, like Principle 7, operates under the operational integrity constraints summarized in Section 4 and is currently a v.5 design-stage framework, not yet implemented in shipped MILO code.

Operational specification is forthcoming work. Principles 7 and 8 are introduced here as architectural commitments; their operational specification — including the baseline-establishment window for Principle 7, the recalibration cadence under operator life-events (illness, role change, circadian variation), the dose-response of the precision perturbation in Principle 8, and the drift-control mechanism for the perturbation magnitude over time — is the subject of forthcoming work tied to the v.5 development program. The principles articulate the design target; the operational validation against measurable outcomes is empirical work that follows from the principles, not work that precedes them.

4. Operational Integrity as Architectural, Not Policy-Level

The eight principles describe the design space of viable adaptive AI orchestration. The operator-layer principles in Sections 3.7 and 3.8 admit deployment patterns that, without operational integrity constraints, would risk surveillance, coercion, or productivity enforcement. The architecture proposed here requires that those constraints be *structural*, not merely policy-level — implemented as enforceable code-level constraints in deployment builds and bound to the architectural design specification. The eight integrity constraints — *no coercion ever*, *individual baseline only*, *no surveillance architecture*, *operator authority is the invariant*, *operational transparency*, *data sovereignty*, *override always available*, and *independent oversight* — must operate as enforceable safeguards, not promises. Deployment patterns that cannot satisfy all eight are outside the permitted use boundary of the architecture by design.

5. Implications and Discussion

Taken together, the eight principles constrain the design space of viable adaptive AI orchestration more tightly than the field currently acknowledges. The first six establish that adaptive AI is bounded by physical, informational, control-theoretic, and statistical constraints that have been known for between sixty and one hundred and seventy years. The seventh and eighth extend the framework to the operator-cognitive performance layer in high-consequence industrial environments, where human decision context can become a material source of system variance — and where the architectural form of the intervention is precision-targeted rather than population-averaged, perturbation-based rather than override-based.

The framework is consistent with, and complementary to, several prior contributions. Friston's Free Energy Principle[3] provides a unified explanation of self-organizing systems at a level above architectural design; the eight principles operate one level below, as architectural constraints. Beer's Viable System Model [4] applies cybernetic feedback to recursive viable systems, especially organizations; the eight principles apply comparable viability discipline at the software-orchestration scale. Thermodynamics-inspired AI work [5] shows how thermodynamic concepts can illuminate black-box AI behavior; the present framework treats entropy as architectural diagnostic. Antifragility[10] provides the payoff-distribution analysis; the present framework treats antifragility as an architectural property of the adaptive-learning loop bounded by operator-authority constraints. The synthesis here is intended to complement, not displace, those frameworks.

The framework leaves substantive work to subsequent contributions on adjacent architectural concerns: the application of the entropy diagnostic to multi-source thermal capture for cryptographic seeding, the application of pre-execution gating to latency-aware authentication in industrial control environments, the architectural form of supervisory primacy in human-in-the-loop AI orchestration for high-consequence domains, and the synthesis of antifragility, viability, and resilience into a unified design ethos — each developed in related work by the author.

5.1 Limitations

This is a perspective paper synthesizing eight principles. Five specific limitations bound the present contribution: (i) the mapping of each established law (Principles 1–6) to a specific orchestrator-level architectural choice is the author's contribution and is subject to further validation; alternative mappings are plausible and not surveyed here; (ii) the relationship of Ashby's Law of Requisite Variety[6] to a fleet of specialist agents is a strong-form analogy rather than a derived result; (iii) Principle 6 (Power-Law Distribution) is an empirical regularity rather than a derived law in the strict sense, as noted explicitly in §3.6; (iv) Principles 7 and 8 — *Individual-Baseline Variance Modeling* and *Precision Perturbation Without Variance Compression* — are original frameworks pending empirical validation and are explicitly flagged as design-stage in §3.7 and §3.8 (their operational specification, including baseline-establishment windows and dose-response of the perturbation, is forthcoming work); (v) the synthesis does not claim a unified theory of intelligence and is explicitly distinguished from Friston's Free Energy Principle[3] in this respect.

6. Conclusion

This paper has identified eight structural principles for adaptive AI architecture. Six are established physical and informational laws applied as architectural design constraints; two are original frameworks proposed by the author for the operator-cognitive performance layer. The principles operate as design-time constraints on architectural choices, not as a unified theory of intelligence. They constrain the design space of viable adaptive AI orchestration more tightly than current field discourse acknowledges. The contribution is at the level of engineering design — what an adaptive AI orchestration system must do architecturally to satisfy the constraints — and is illustrated using MILO, the author's working implementation[1], submitted under the U.S. Department of Energy's Genesis Mission[2]. The principles are intended to be falsifiable, applicable, and consistent with the broader literature on adaptive systems, cybernetics, and the thermodynamics of intelligence.

Data Availability

All architectural materials, source manuscripts, the reference implementation, and accompanying figures are openly available at <https://github.com/jmontano1/milo-architecture> and permanently archived at Zenodo (DOI: [10.5281/zenodo.20117025](https://doi.org/10.5281/zenodo.20117025)). No private datasets are referenced; the architectural framework itself is the subject of this paper. Patent rights for the underlying MILO software architecture are reserved; the ~MILO trademark is held under USPTO Serial No. 99706004 (intent-to-use, Class 009).

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Conflict of Interest and Funding Disclosure

The author is the inventor of MILO (patent pending) and the founder of JM Automated Solutions. The eight structural principles articulated in this paper, including the two original frameworks proposed in Sections 3.7 and 3.8, are a contribution from a working development program in which the author retains sole authorship and inventive interest. No external funding was received for the preparation of this manuscript. The author retains all rights to MILO and to the original frameworks introduced herein.